Week 5 · NYC taxi data pipeline







Forecasting NYC taxi demand

How many pickups will each zone get in each hour?









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The data

23.5 M

cleaned trips:
yellow, green and FHV
with a pickup zone

1.15 M

zone-hour rows:
265 zones × 4,344 hours

6 months

Jan - Jun 2026
split by time

Train · Jan - Apr 2026

718,680 rows

Val · May 2026

197,160 rows

Test · June 2026

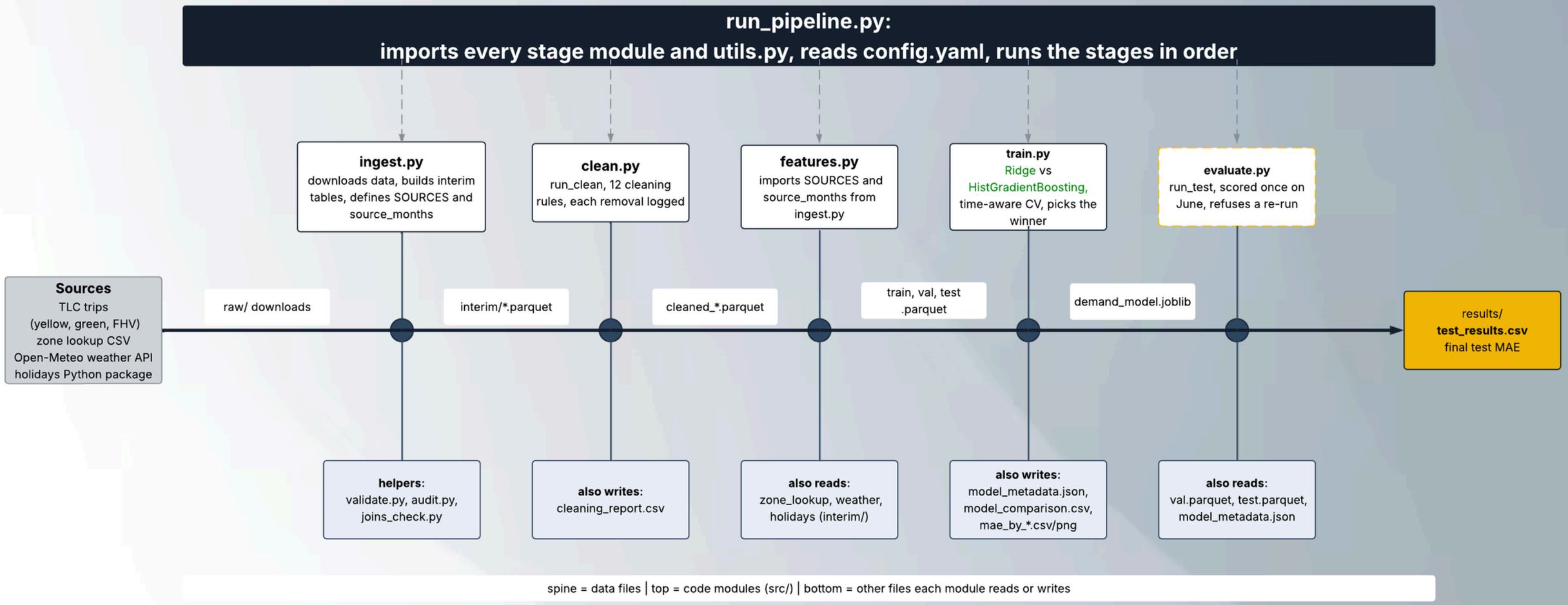
190,800 rows

***And hourly weather (Open-Meteo), NY holidays and the TLC zone lookup.**

How the pipeline is built

Ingest ingest, validate, audit	Download trips, zones, weather, holidays; validate schema, row counts, weather coverage; plan joins	`interim/*.parquet, data_audit.csv`
clean clean.py	12 rules (timestamps, duration, distance, fare, zone), each removal logged; FHV rows without a zone dropped	`cleaned_*.parquet, cleaning_report.csv`
features features.py	Zone-hour aggregation, complete 265 × 4,344 grid, joins with row-count asserts, lags, split by month	`train, val, test.parquet`
train train.py	Sklearn Pipeline per model, 3-fold TimeSeriesSplit, winner chosen by validation MAE	`demand_model.joblib, metadata, MAE tables`
test evaluate.py	Locked June month, scored once; refuses to run a second time	`test_results.csv`

How the files connect



Data and feature decisions

Complete zone-hour grid

A plain groupby drops hours with zero trips and overstates demand, so every zone × hour exists, with 0 where nothing happened.

Weather joined after aggregationc

A per-trip join repeats hourly values millions of times. Every join asserts that the row count did not change.

Lags use only the past

lag_1, lag_24, lag_168 and a rolling mean shifted by one hour. No leakage; the first week is dropped (44,520 rows).

Split by month, not randomly

Train Jan -Apr, validate May, test June. A random split would leak future demand into training.

Quality checks: 12 cleaning rules; 84–92% of FHV rows have no pickup zone and are dropped.

Why gradient boosting

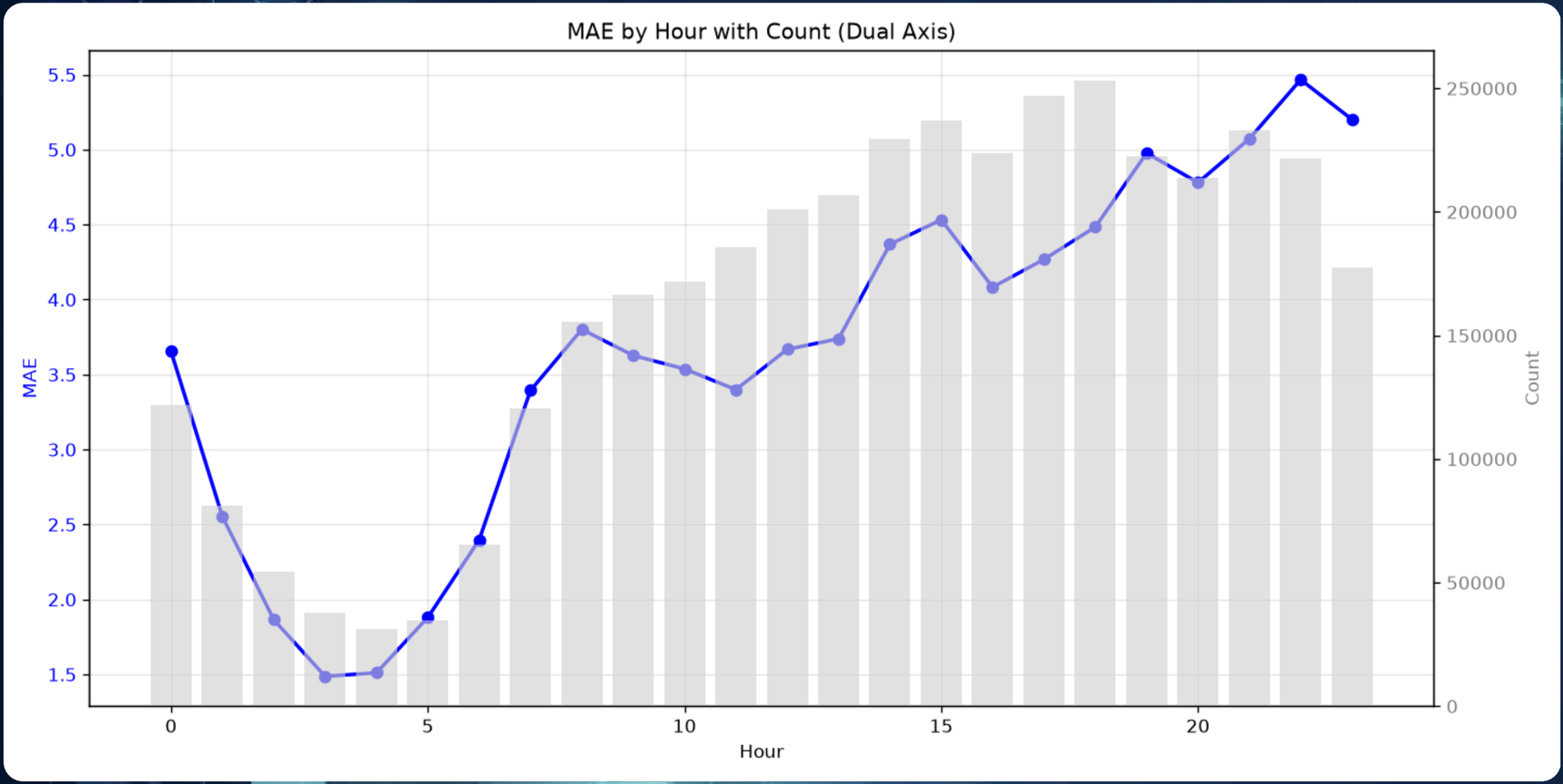
Model	CV MAE	Val MAE
Baseline		5.10
Ridge	4.21 ± 0.33	4.02
Hist Gradient Boosting	3.61 ± 0.14	3.66

More stable: *fold-to-fold std 0.14 versus 0.33.*

Fits the shape of the data: *trees capture hour × zone effects that a linear model needs hand-made interactions for.*

Honest selection: *chosen on validation; the test month was never used for any decision.*

Where it fails: by hour



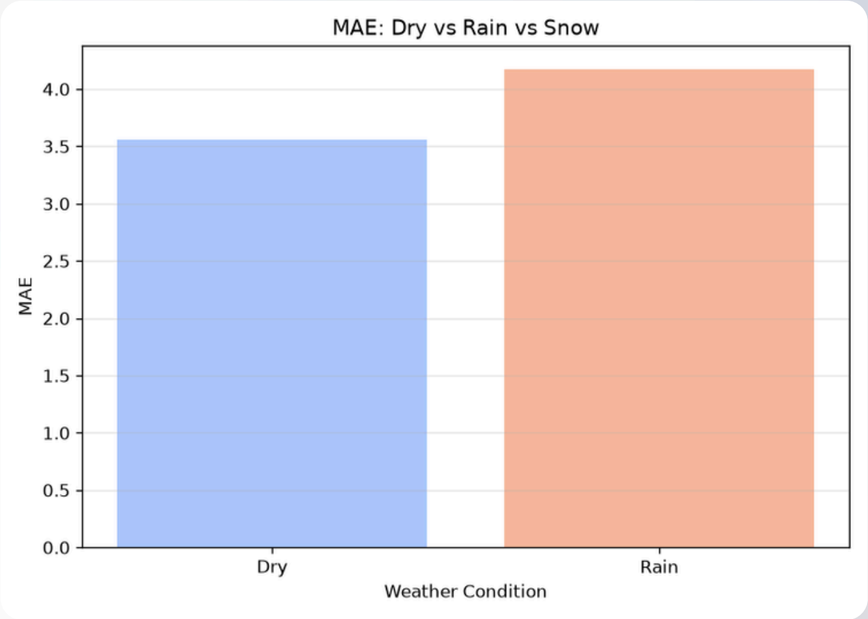
Best at night: MAE 1.5 at 03:00.

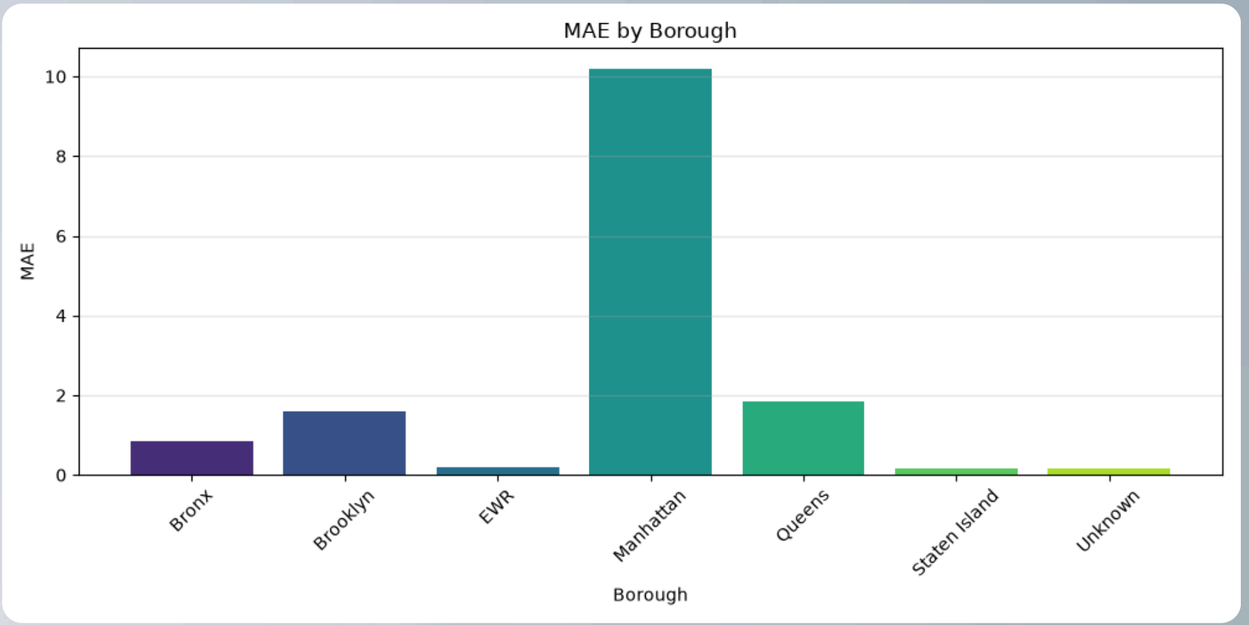
Worst in the evening: 5.5 at 22:00.

Error follows volume: grey bars are trips per hour, and they rise with MAE.

Likely cause: busier hours are larger and noisier, and MAE is absolute.

Where it fails: borough and weather





Rain 4.2 VS Dry 3.6.
**No snow bar: validation has no snowfall.*

Manhattan MAE more than 10, *every other borough under 2. Absolute MAE tracks demand volume.*

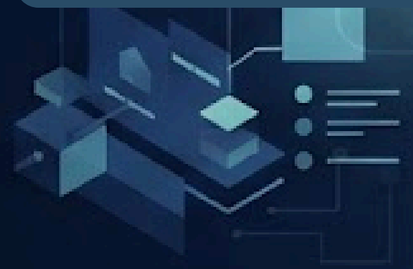
Limits and next steps

Limits

- Six months of data, one test month
- No snow in validation, so snow error is unknown
- Observed weather, one point for the city; real use needs forecasts
- Most FHV trips lack a pickup zone

What next

- Reuse: validation, logging, train and evaluate structure
- Change: config, cleaning rules, features and target
- Next: more months, the holiday flag, weather forecasts







Thank you for your attention!
I will be happy to answer your questions.